FINANCIAL LOAN ANALYSIS

Python-Based Financial Loan Analysis & Business Insights

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1. Introduction

Financial Loan Analysis is a data analysis project developed using Python to examine loan applications, funded amounts, repayments, borrower information, and loan performance.

The main purpose of this project is to analyses financial loan data and generate meaningful insights that can help understand lending activities, repayment performance, and borrower characteristics.

Using Python libraries such as Pandas, Matplotlib, and Seaborn, the dataset is cleaned, transformed, analyzed, and visualized through different charts and summary reports.

The project focuses on three main areas: loan performance, key financial indicators, and borrower-based analysis.

2. Project Objectives

The main objectives of this project are:

- Analyze overall loan applications, funded amounts, and received amounts.
- Calculate important financial KPIs, including average interest rate and debt-to-income ratio.
- Compare good loans and bad loans based on loan status.
- Analyze monthly lending trends and Month-over-Month (MoM) changes.
- Understand loan distribution across different states, loan terms, employment lengths, loan purposes, and home ownership categories.
- Create visualizations to present financial information clearly and support data-driven decisions.

3. Dataset Description

The project uses a financial loan dataset containing 38,576 records and 24 original columns.

The dataset includes information about loan applications, borrowers, loan amounts, repayments, interest rates, and loan status.

Some important columns include:

Columns	**Description**
id	Unique loan application ID
address_state	Borrower's state
application_type	Type of loan application
emp_length	Borrower's employment length

emp_title	Borrower's employment title
grade	Loan grade
home_ownership	Borrower's home ownership status
issue_date	Loan issue date
loan_status	Current status of the loan
purpose	Purpose of the loan
term	Loan repayment term
annual_income	Borrower's annual income
dti	Debt-to-income ratio
int_rate	Loan interest rate
loan_amount	Amount funded for the loan
total_payment	Total amount received

4. Tools and Technologies

The following tools and Python libraries were used:

- **Python:** Main programming language used for data analysis.
- **Jupyter Notebook:** Development environment for writing and executing Python code.
- **Pandas:** Used for data loading, cleaning, transformation, grouping, aggregation, and KPI calculations.
- **Matplotlib:** Used to create charts and customise visualisations.
- **Seaborn:** Used to create statistical and comparative visualisations.

5. Project Implementation

5.1 Data Loading and Inspection

The dataset was imported into a Pandas DataFrame using read_excel().

Initial inspection was performed using:

- head() to view sample records.
- info() to understand column names, data types, and non-null values.
- columns to inspect the available dataset fields.

This helped understand the dataset structure before performing further analysis.

5.2 Data Cleaning

Data cleaning was performed to improve the quality and consistency of the dataset.

The following operations were carried out:

- Checked missing values using isnull().sum().
- Checked duplicate records using duplicated().
- Replaced missing values in the emp_title column with Unknown Title.
- Checked missing values again to confirm that the replacement was applied.

The dataset contained 1,438 missing values in the employee title column and no duplicate rows.

After replacing the missing employee titles, the dataset contained 38,576 records with no missing values.

5.3 Feature Engineering

Feature engineering was performed to extract additional information from the existing date column.

Two new columns were created:

- month_num: Extracted the numerical month from issue_date.
- issue_month: Extracted the month name from issue_date.

These columns were used to group loan applications by month and analyse monthly lending trends.

5.4 Summary KPI Analysis

Important financial KPIs were calculated to understand the overall performance of the loan portfolio.

The following KPIs were analyzed:

1. Total Loan Applications

Calculated the total number of loan applications using the loan application ID.

2. Total Funded Amount

Calculated the total amount of money disbursed as loans using loan_amount.

3. Total Received Amount

Calculated the total amount received from borrowers using total_payment.

4. Average Interest Rate

Calculated the average interest rate across all loan applications.

5. Average Debt-to-Income Ratio (DTI)

Calculated the average DTI to understand the financial position of borrowers.

For these KPIs, the project also calculated:

- Month-to-Date (MTD) values.
- Previous Month-to-Date (PMTD) values.
- Month-over-Month (MoM) percentage changes.

The following formula was used to calculate percentage growth:

MoM Growth (%) = ((Current Month Value - Previous Month Value) / Previous Month Value) × 100

5.5 Good Loan and Bad Loan Analysis

Loans were categorized into two groups based on their status.

Good Loans:

- Current
- Fully Paid

Bad Loans:

- Charged Off

The analysis calculated the following metrics for both categories:

- Total loan applications.
- Percentage of loan applications.
- Total funded amount.
- Total received amount.

This analysis helps compare loan performance and understand the difference between successfully performing loans and charged-off loans.

5.6 Loan Status Grid View

A summary table was created by grouping the dataset using loan_status.

The table includes:

- Total Loan Applications.
- Total Funded Amount.
- Total Received Amount.
- Average Interest Rate.
- Average DTI.
- MTD Funded Amount.
- MTD Received Amount.

Custom Python functions were also created to format currency and percentage values for better readability.

6. Exploratory Data Analysis and Visualization

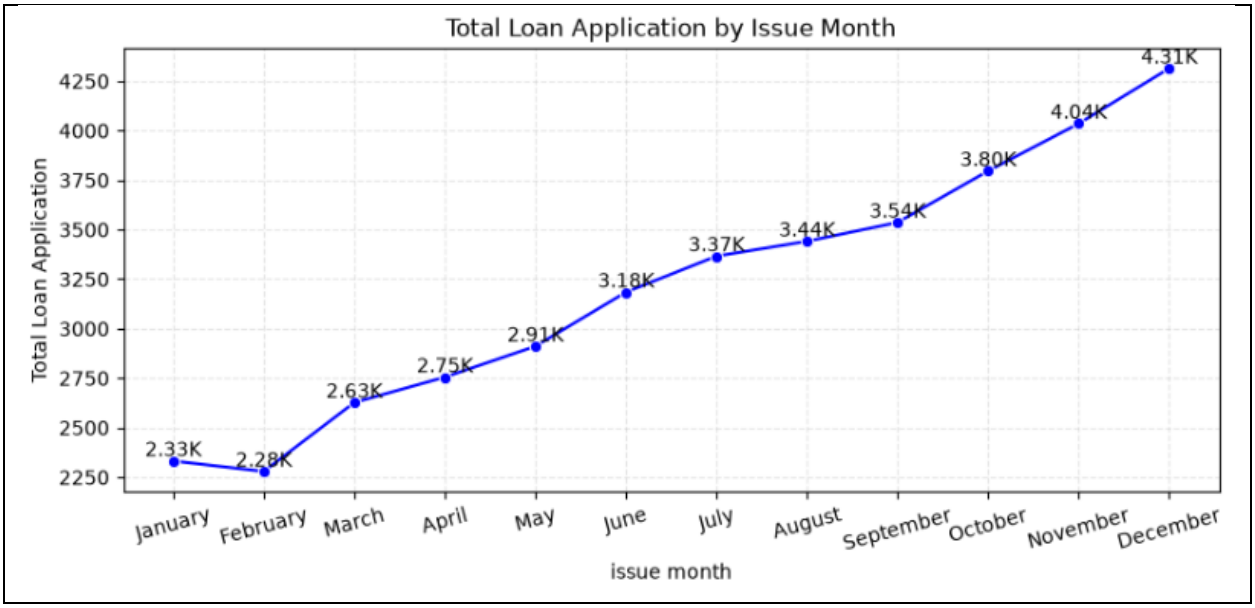
Different analytical categories were examined using Pandas grouping and aggregation techniques. Matplotlib and Seaborn were used to create visualizations.

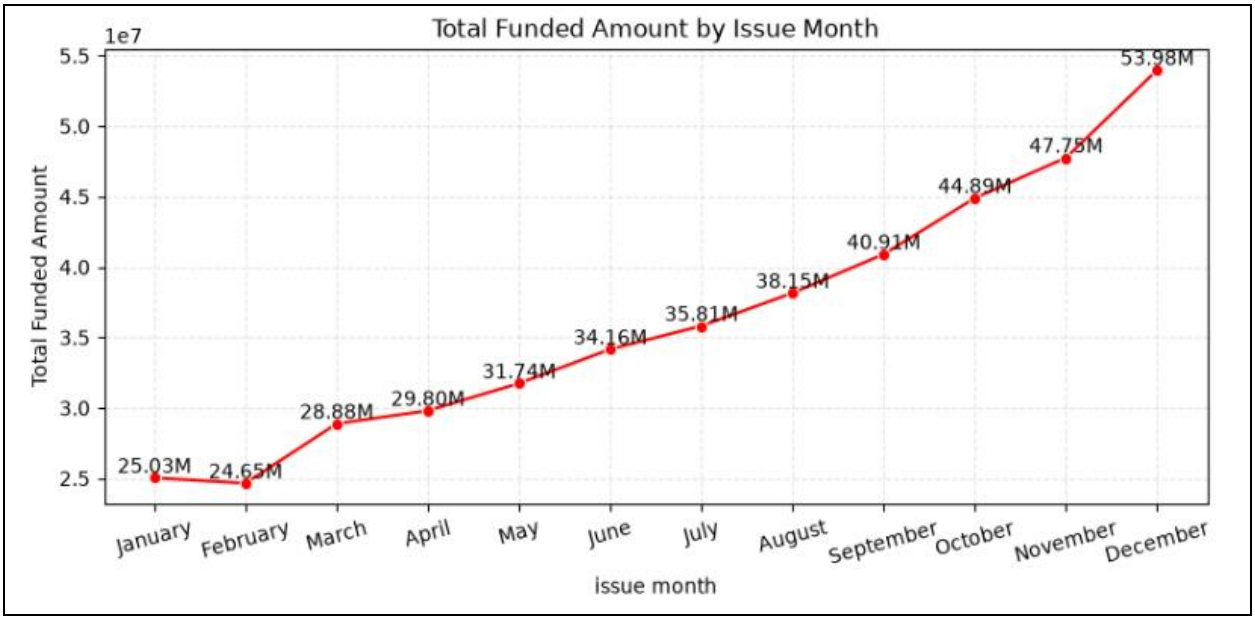
6.1 Monthly Trends by Issue Date

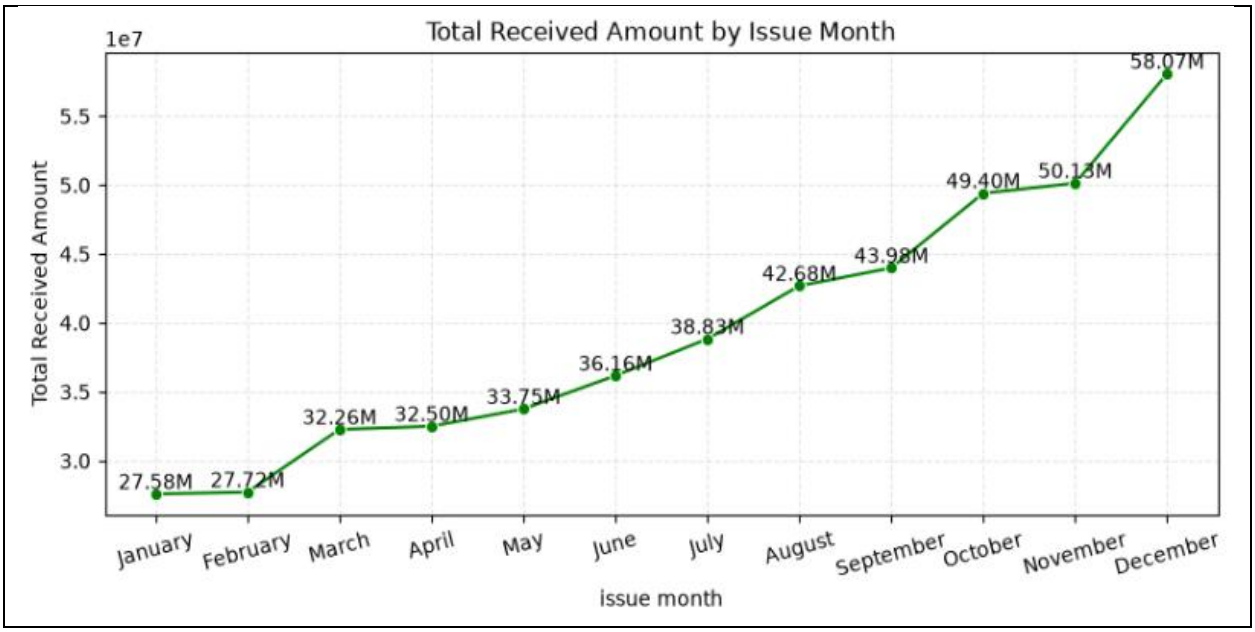
Business Insight: Loan applications, funded amounts, and received amounts generally increased towards the end of the year. December recorded the highest lending activity, while February had the lowest number of applications.

Issue Month	Total Loan Applications	Total Funded Amount	Total Received Amount
January	2,332	$25,031,650	$27,578,836
February	2,279	$24,647,825	$27,717,745
March	2,627	$28,875,700	$32,264,400
April	2,755	$29,800,800	$32,495,533
May	2,911	$31,738,350	$33,750,523
June	3,184	$34,161,475	$36,164,533
July	3,366	$35,813,900	$38,827,220
August	3,441	$38,149,600	$42,682,218
September	3,536	$40,907,725	$43,983,948
October	3,796	$44,893,800	$49,399,567
November	4,035	$47,754,825	$50,132,030

December	4,314	$53,981,425	$58,074,380



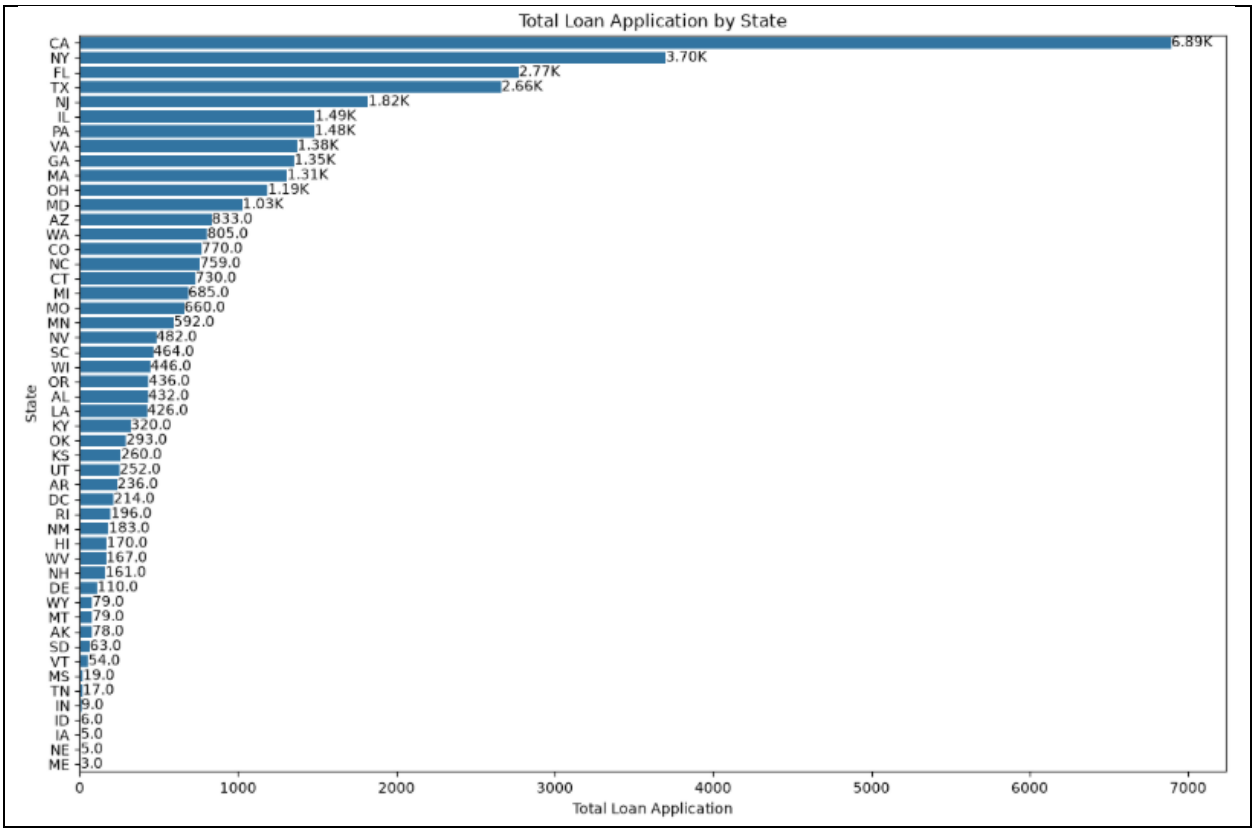


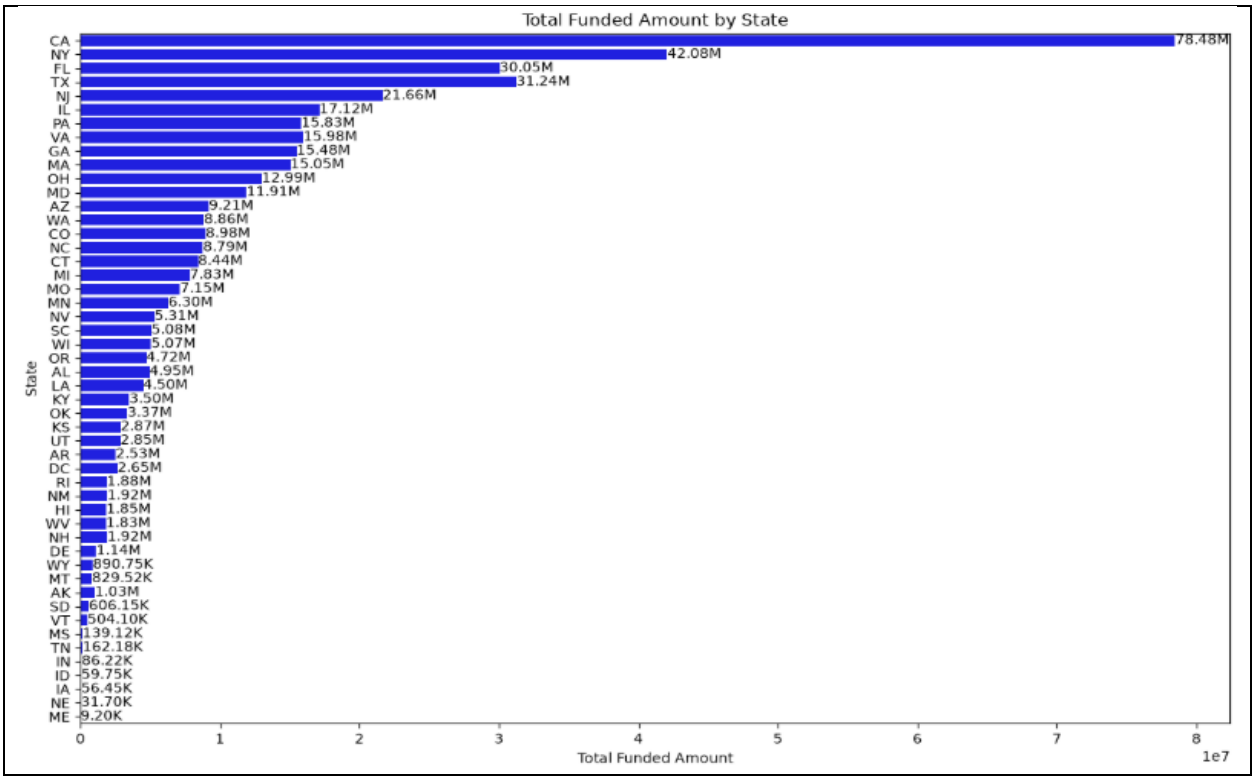


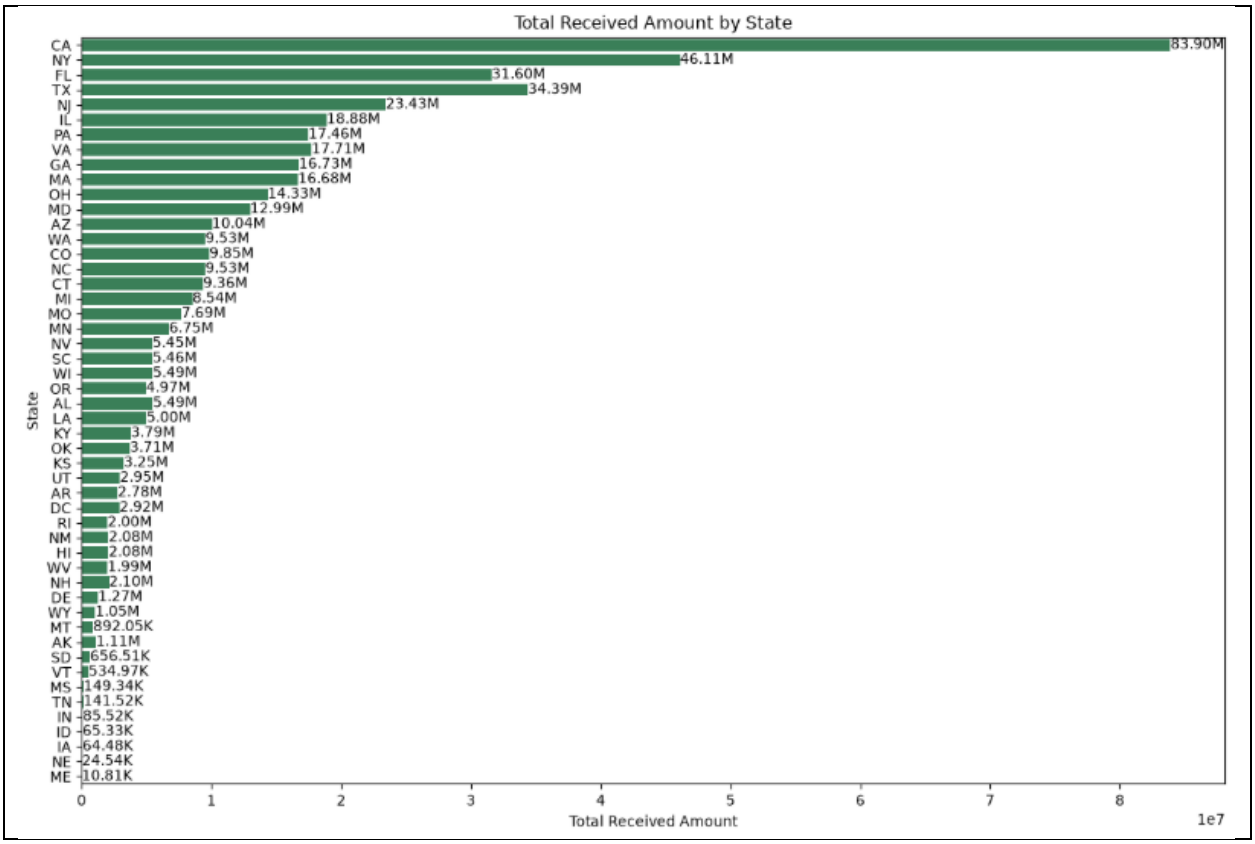
6.2 Regional Analysis by State

Business Insight: The analysis compares loan applications, funded amounts, and received amounts across different states. California recorded 6,894 applications and the highest figures among the five states displayed in the notebook output.

State	Total Loan Application	Total Funded Amount	Total Received Amount
AK	78	$1,031,800	$1,108,570
AL	432	$4,949,225	$5,492,272
AR	236	$2,529,700	$2,777,875
AZ	833	$9,206,000	$10,041,986
CA	6,894	$78,484,125	$83,901,234


Total Loan Application by State
CA 6.89K
NY 3.70K
FL 2.77K
TX 2.66K
NJ 1.82K
IL 1.49K
PA 1.48K
VA 1.38K
GA 1.35K
MA 1.31K
OH 1.19K
MD 1.03K
AZ 833.0
WA 805.0
CO 770.0
NC 759.0
CT 730.0
MI 685.0
MO 660.0
MN 592.0
NV 482.0
SC 464.0
WI 446.0
OR 436.0
AL 432.0
LA 426.0
KY 320.0
OK 293.0
KS 260.0
UT 252.0
AR 236.0
DC 214.0
RI 196.0
NM 183.0
HI 170.0
WV 167.0
NH 161.0
DE 110.0
WY 79.0
MT 79.0
AK 78.0
SD 63.0
VT 54.0
MS 19.0
TN 17.0
IN 9.0
ID 6.0
IA 5.0
NE 5.0
ME 3.0
State
0
1000
2000
3000
4000
5000
6000
7000
Total Loan Application

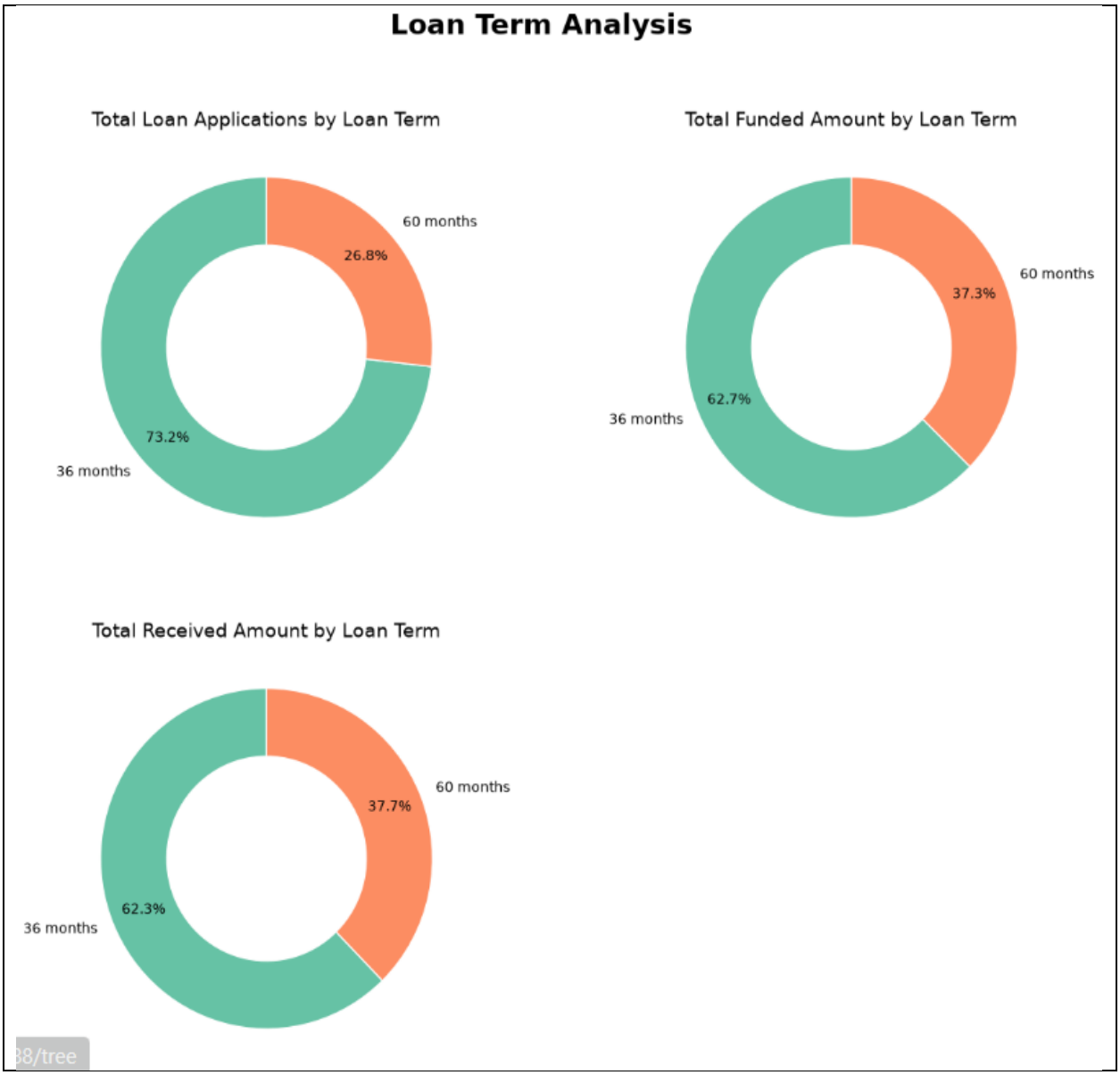

Total Funded Amount by State
CA 78.48M
NY 42.08M
FL 30.05M
TX 31.24M
NJ 21.66M
IL 17.12M
PA 15.83M
VA 15.98M
GA 15.48M
MA 15.05M
OH 12.99M
MD 11.91M
AZ 9.21M
WA 8.86M
CO 8.98M
NC 8.79M
CT 8.44M
MI 7.83M
MO 7.15M
MN 6.30M
NV 5.31M
SC 5.08M
WI 5.07M
OR 4.72M
AL 4.95M
LA 4.50M
KY 3.50M
OK 3.37M
KS 2.87M
UT 2.85M
AR 2.53M
DC 2.65M
RI 1.88M
NM 1.92M
HI 1.85M
WV 1.83M
NH 1.92M
DE 1.14M
WY 890.75K
MT 829.52K
AK 1.03M
SD 606.15K
VT 504.10K
MS 139.12K
TN 162.18K
IN 86.22K
ID 59.75K
IA 56.45K
NE 31.70K
ME 9.20K
State
0
1
2
3
4
5
6
7
8
Total Funded Amount
1e7



6.3 Loan Term Analysis

Business Insight: Most borrowers selected the 36-month loan term, accounting for 28,237 applications. This term also recorded higher funded and received amounts than the 60-month term.

Loan	Total Loan Application	Total Funded Amount	Total Received Amount
36 months	28,237	$273,041,225	$294,709,458
60 months	10,339	$162,715,850	$178,361,475

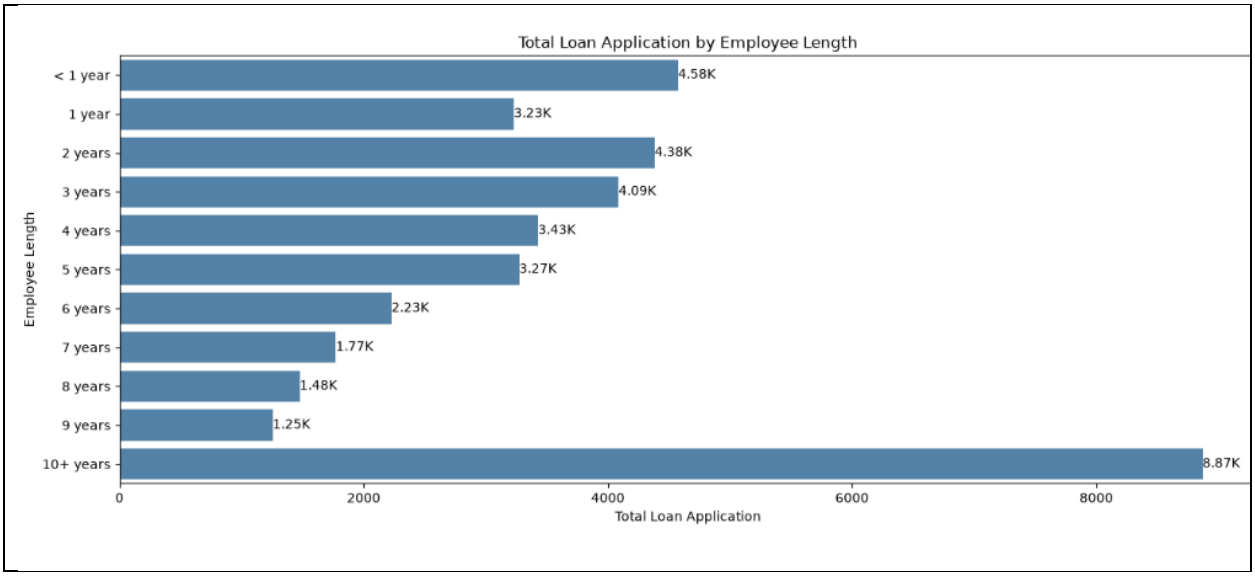


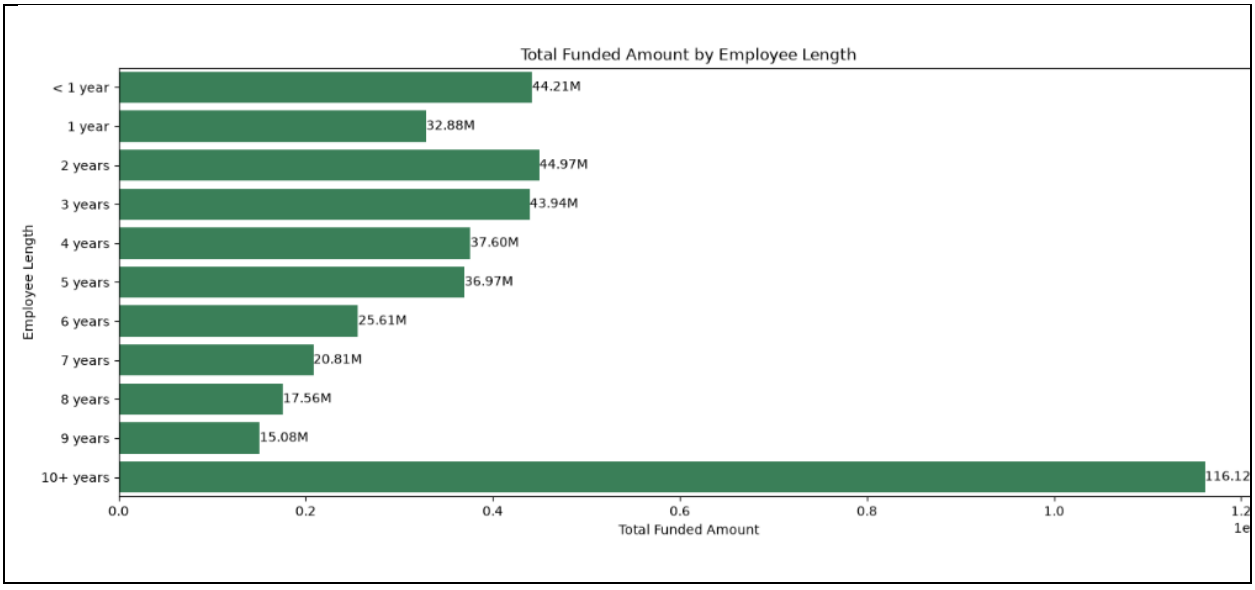
6.4 Employee Length Analysis

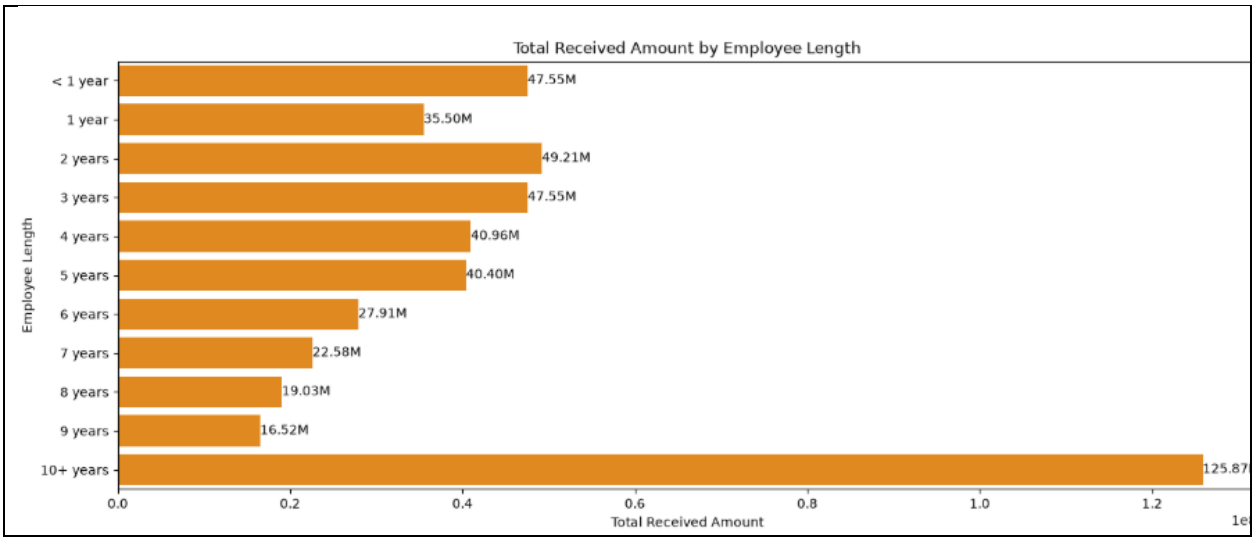
Business Insight: Borrowers with 10 or more years of employment recorded the highest number of applications, funded amount, and received amount. This group represents the largest employment-length category in the dataset.

Employee Length	Total Loan Applications	Total Funded Amount	Total Received Amount
Less than 1 year	4,575	$44,210,625	$47,545,011
1 year	3,229	$32,883,125	$35,498,348

2 years	4,382	$44,967,975	$49,206,961
3 years	4,088	$43,937,850	$47,551,832
4 years	3,428	$37,600,375	$40,964,850
5 years	3,273	$36,973,625	$40,397,571
6 years	2,228	$25,612,650	$27,908,658
7 years	1,772	$20,811,725	$22,584,136
8 years	1,476	$17,558,950	$19,025,777
9 years	1,255	$15,084,225	$16,516,173
10+ years	8,870	$116,115,950	$125,871,616



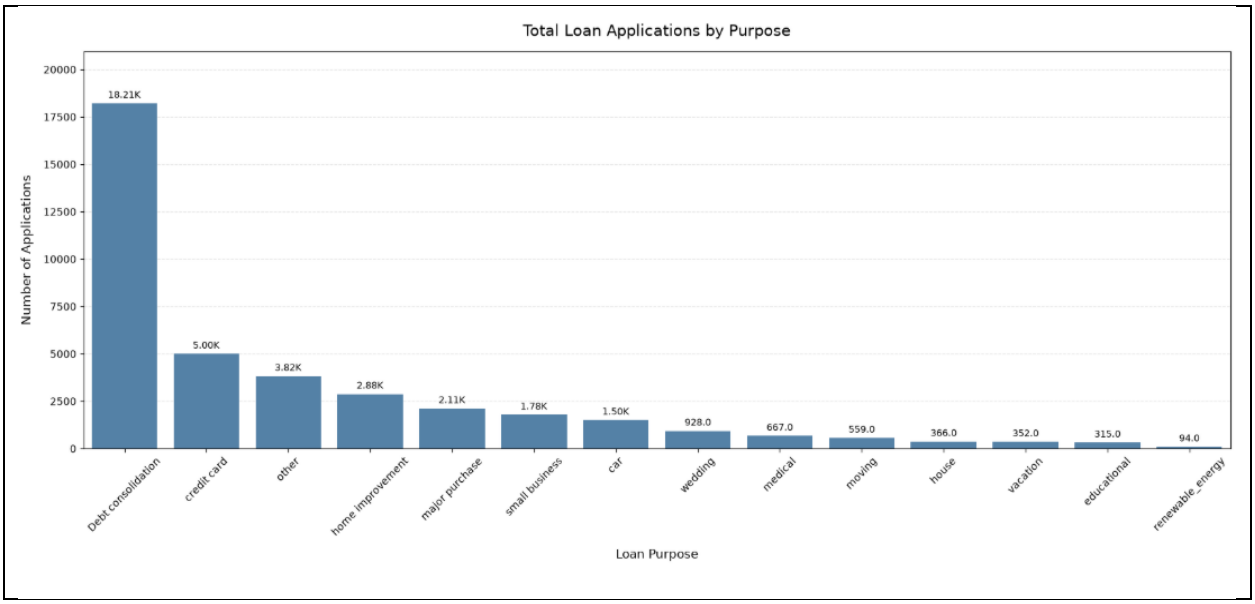


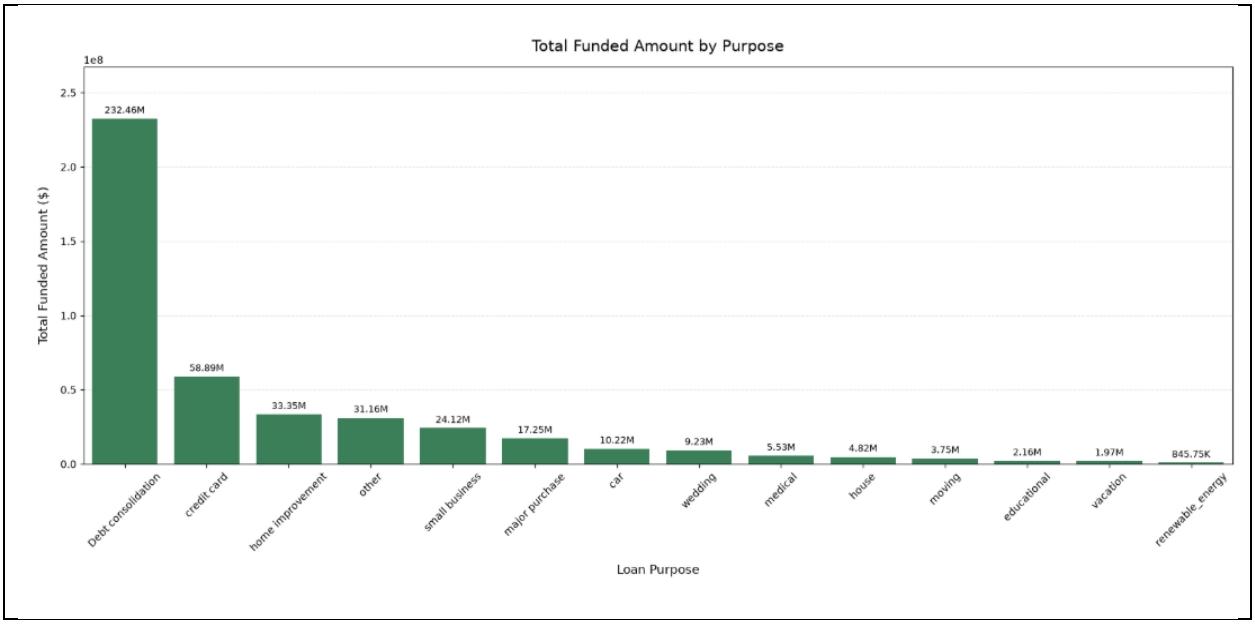


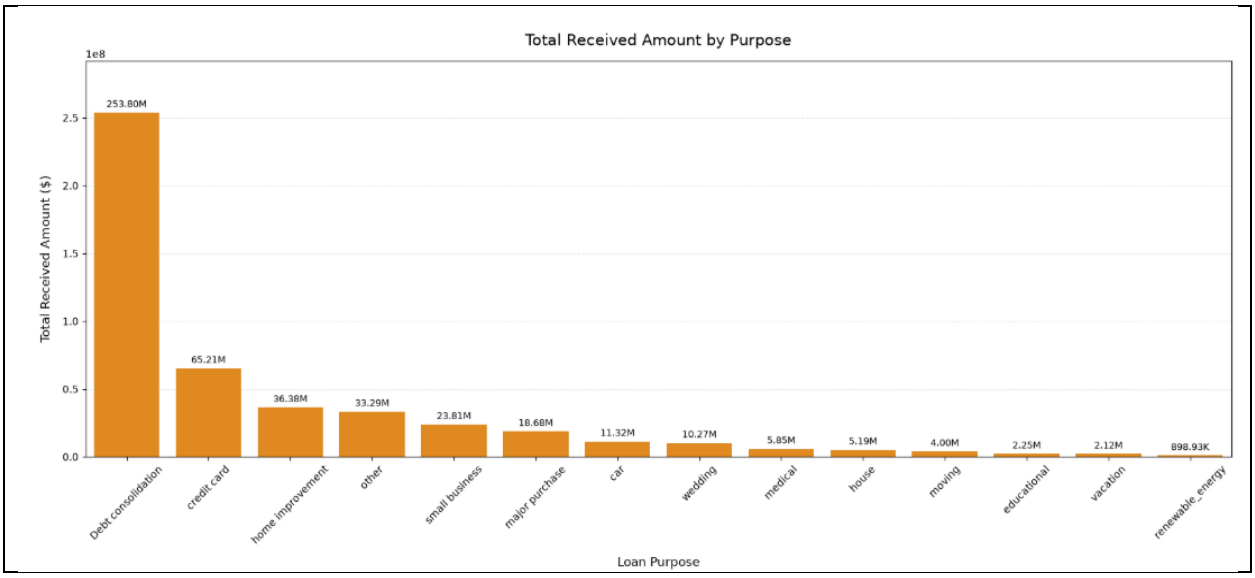
6.5 Loan Purpose Analysis

Business Insight: Debt consolidation was the most common loan purpose, with 18,214 applications and the highest funded and received amounts. Credit card payments and other purposes were also significant categories.

Loan Purpose	Total Loan Applications	Total Funded Amount	Total Received Amount
Debt consolidation	18,214	$232,459,675	$253,801,871
Car	1,497	$10,223,575	$11,324,914
Credit card	4,998	$58,885,175	$65,214,084
Educational	315	$2,161,650	$2,248,380
Home improvement	2,876	$33,350,775	$36,380,930
House	366	$4,824,925	$5,185,538
Major purchase	2,110	$17,251,600	$18,676,927
Medical	667	$5,533,225	$5,851,372
Moving	559	$3,748,125	$3,999,899
Other	3,824	$31,155,750	$33,289,676
Renewable energy	94	$845,750	$898,931
Small business	1,776	$24,123,100	$23,814,817
Vacation	352	$1,967,950	$2,116,738
Wedding	928	$9,225,800	$10,266,856


Total Loan Applications by Purpose
Number of Applications
Loan Purpose
18.21K
5.00K
3.82K
2.88K
2.11K
1.78K
1.50K
928.0
667.0
559.0
366.0
352.0
315.0
94.0
Debt consolidation
credit card
other
home improvement
major purchase
small business
car
wedding
medical
moving
house
vacation
educational
renewable_energy

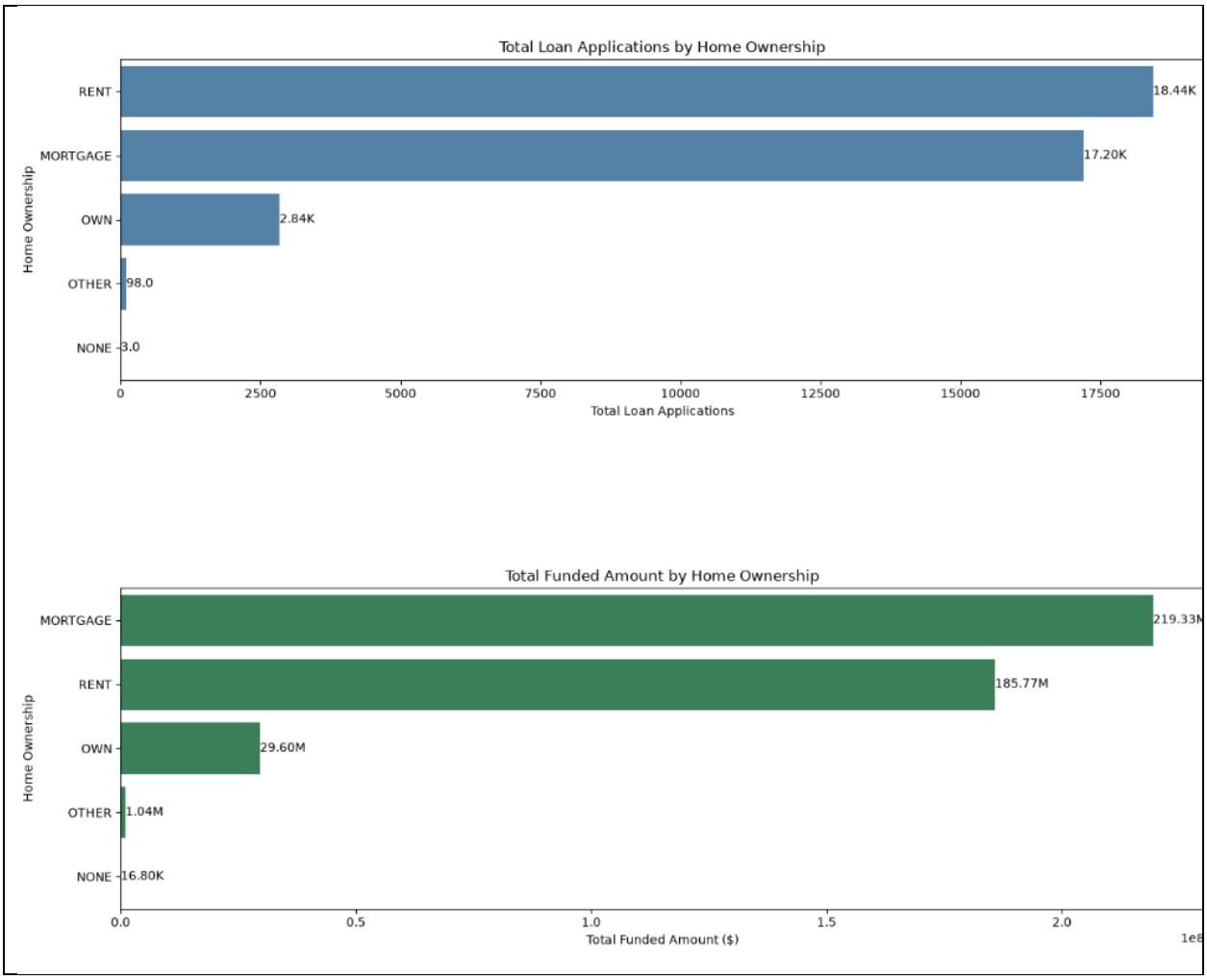

Total Funded Amount by Purpose
1e8
Total Funded Amount ($)
Loan Purpose
232.46M
58.89M
33.35M
31.16M
24.12M
17.25M
10.22M
9.23M
5.53M
4.82M
3.75M
2.16M
1.97M
845.75K
Debt consolidation
credit card
home improvement
other
small business
major purchase
car
wedding
medical
house
moving
educational
vacation
renewable_energy

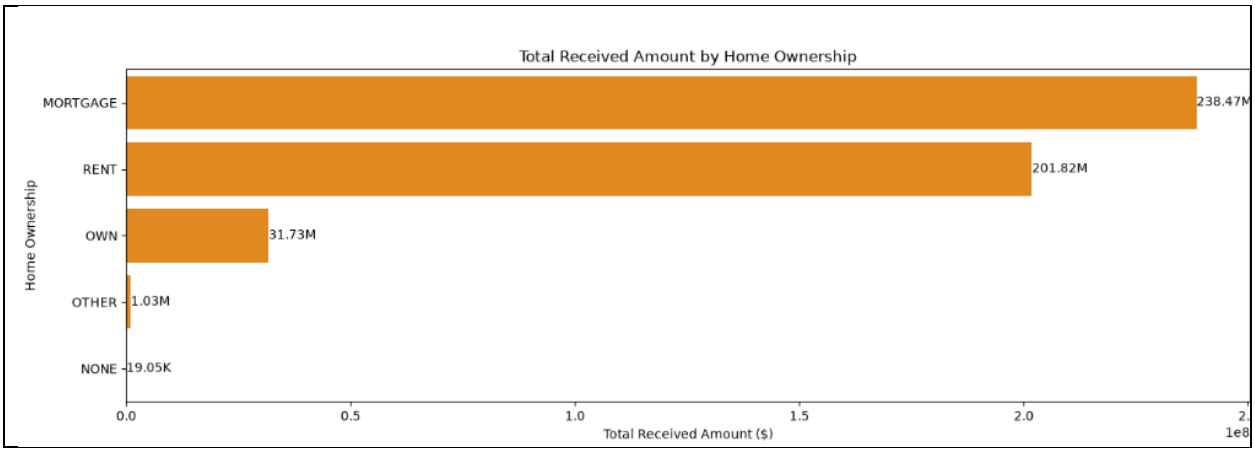


6.6 Home Ownership Analysis

Business Insight: Renters submitted the highest number of loan applications, while borrowers with mortgages recorded the highest funded and received amounts. This shows differences in loan activity across home ownership categories.

Home Ownership	Total Loan Applications	Total Funded Amount	Total Received Amount
Mortgage	17,198	$219,329,150	$238,474,438
None	3	$16,800	$19,053
Other	98	$1,044,975	$1,025,257
Own	2,838	$29,597,675	$31,729,129
Rent	18,439	$185,768,475	$201,823,056


Total Loan Applications by Home Ownership
RENT
MORTGAGE
OWN
OTHER
NONE
Home Ownership
18.44K
17.20K
2.84K
98.0
3.0
0
2500
5000
7500
10000
12500
15000
17500
Total Loan Applications
Total Funded Amount by Home Ownership
MORTGAGE
RENT
OWN
OTHER
NONE
Home Ownership
219.33M
185.77M
29.60M
1.04M
16.80K
0.0
0.5
1.0
1.5
2.0
Total Funded Amount ($)
1e8


Total Received Amount by Home Ownership
MORTGAGE
RENT
OWN
OTHER
NONE
Home Ownership
238.47M
201.82M
31.73M
1.03M
19.05K
0.0
0.5
1.0
1.5
2.0
Total Received Amount ($)
1e8

7. Key Results and Findings

Based on the analysis performed, the following results were obtained:

Metric	Results
Total Loan Applications	38,576
Total Funded Amount	$435,757,075
Total Received Amount	$473,070,933
Average Interest Rate	12.05%
Average DTI	13.33%
Good Loan Applications	33,243
Good Loan Percentage	86.18%
Bad Loan Applications	5,333
Bad Loan Percentage	13.82%

Additional observations:

- December recorded the highest number of loan applications, with 4,314 applications.
- Debt consolidation was the most common loan purpose, with 18,214 applications.
- The 36-month loan term had 28,237 applications, compared with 10,339 for the 60-month term.
- Borrowers with 10+ years of employment recorded the highest number of applications among employment-length categories.
- Borrowers with rented homes accounted for 18,439 applications, while mortgage holders accounted for 17,198 applications.

These findings provide an overview of loan distribution, borrower characteristics, and lending performance within the dataset.

8. Conclusion

The Financial Loan Analysis project demonstrates how Python can be used to analyse financial datasets and generate meaningful business insights.

Using Pandas, the dataset was inspected, cleaned, transformed, and analyzed to calculate important financial KPIs and identify lending patterns across different loan statuses, months, states, repayment terms, employment lengths, loan purposes, and home ownership categories.

Matplotlib and Seaborn were used to create visualizations that make analytical results easier to understand and compare.

Through this project, practical experience was gained in data cleaning, feature engineering, data aggregation, KPI calculation, exploratory data analysis (EDA), and data visualization.

Overall, the project demonstrates the application of Python-based data analysis techniques to a financial dataset and the ability to convert raw data into structured information that supports better business understanding and decision-making.